Thesauros — AI Layer over PSO

Audience: CEO / CTO **Status:** Product concept **Scope:** Architecture & governance (not implementation)

1. Executive summary

Thesauros is a yield vault that allocates capital across lending protocols using PSO. An AI layer sits above the optimizer: it continuously rebuilds *how* allocation is searched — market regime, risk penalties, and swarm parameters — using the vault's own history and external market signals. PSO still computes percentage weights. Capital movement remains with an operator or Safe multisig.

> **Positioning:** AI manages allocation strategy and PSO inputs — not signing keys.

Differentiation vs a plain APR rebalancer

- PSO input is a risk-adjusted forecast, not a raw on-chain deposit rate
- Optimizer settings are retuned from data and backtests, not hardcoded forever
- Every proposal is explainable and measured against a baseline
- Capital execution is explicitly human- or multisig-gated

2. Core idea

Component	Role
Data	Vault history (rates, TVL, rebalances, backtests) + external market features
AI layer	Forecast, risk scoring, regime/meta tuning, optional explanation
PSO	Searches allocation percentages under AI-provided objective and parameters
Governance	Operator / Safe confirms; on-chain vault executes

AI proposes and validates via backtests. AI does not submit transactions.

3. Architecture — three layers

Layer 1 — Forecast + Risk (ML)

Two focused models trained on vault data and market features:

- **Forecast** — expected APR per provider over a short horizon
- **Risk** — instability / drawdown likelihood of the current rate signal

PSO receives a **risk-adjusted signal** aligned to the vault profile (e.g. conservative / pro), not the raw deposit rate.

Layer 2 — Meta / regime + PSO tuning (ML + offline backtests)

A meta model or scheduled offline loop selects market regime, updates PSO hyperparameters, and adjusts provider weights / penalties. Production PSO remains in place: the AI layer changes configuration and objective, it does not replace the swarm with a black-box allocator. Feedback loop: new on-chain outcomes → retrain / re-run backtests → updated coefficients.

Layer 3 — Advisor / explain (optional LLM, non-custodial)

Formats model outputs and the rebalance plan into a human-readable proposal: what changed, why, expected uplift. Does **not** approve capital moves and does **not** send transactions. Leaves a clear trail: AI proposal vs operator/Safe confirmation.

4. Operating principle

```
[Vault DB + market]
        ├──► Forecast model ──┐
        ├──► Risk model ──────┼──► Risk-adjusted signal ──┐
        └──► Meta + backtests ──► Regime & PSO config ────┼──► PSO → allocation %
                                                          │
                                                          ▼
                                                    Rebalance plan
                                                          │
                                                          ▼
                                               Advisor (why + uplift)
                                                          │
                                                          ▼
                                               Operator / Safe confirm
                                                          │
                                                          ▼
                                               On-chain vault ──► DB (learn loop)
```

1. From vault and market data, compute APR forecast and risk per provider.
2. Meta loop selects regime and retunes PSO settings via backtests.
3. PSO searches weights on the adjusted signal, not raw rate.
4. Advisor packages the proposal and rationale.
5. Operator/Safe confirms → on-chain execution → outcome returns to the database for the next learning cycle.

5. Messaging

External	AI manages allocation strategy. Capital execution stays with human control or multisig.
Internal (accurate)	AI manages strategy inputs and PSO configuration. It does not hold or use signing keys.
Proof points	APR uplift vs baselines (Aave-only / legacy PSO); decision log; explicit human-in-the-loop

6. Governance boundary

AI does	AI does not
Forecast and risk scoring	Sign or broadcast transactions
Regime selection and PSO tuning	Final capital decisions
Proposal + explanation	Bypass RiskGuard / Safe

Learn from DB, market, and backtests	Act as autonomous custodian

7. Out of scope

- Model algorithm selection and training pipeline
- Service topology and code changes in the rebalance engine
- Product brand name for the AI layer
- Fully autonomous capital execution

8. Short brief

Thesauros is a yield vault with an AI layer over PSO. Models trained on vault history and market data rebuild the allocation objective and swarm parameters; PSO computes percentages; an operator or Safe moves capital. AI proposes and validates with backtests — it does not sign transactions.

Forecast + Risk → Meta → Advisor | Strategy, not keys | Uplift + decision log + HITL